Lec-38

04/04/24 12:38 PM

261. Graph Valid Tree

Given `n` nodes labeled from `0` to `n-1` and a list of undirected edges (each edge is a pair of nodes), write a function to check whether these edges make up a valid tree.

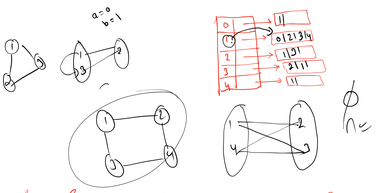
Example 1:

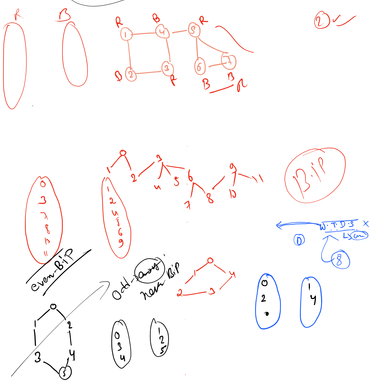
```
Input: n = 5, and edges = [[0,1], [0,2], [0,3], [1,4]]
Output: true
```

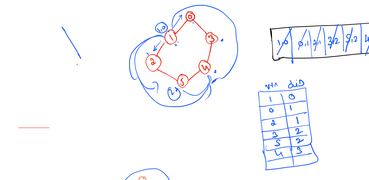
Example 2:

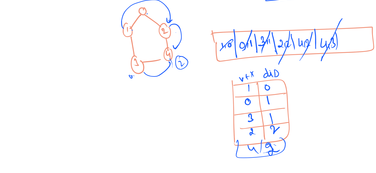
```
Input: n = 5, and edges = [[0,1], [1,2], [2,3], [1,3], [1,4]]
Output: false
```

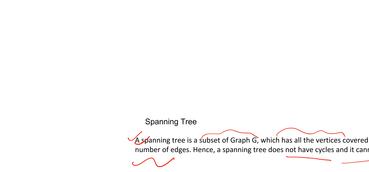
Note: you can assume that no duplicate edges will appear in `edges`. Since all edges are undirected, `[0,1]` is the same as `[1,0]` and thus will not appear together in `edges`.

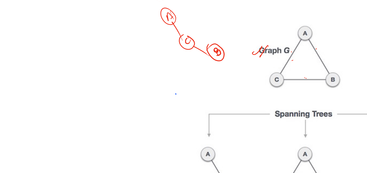


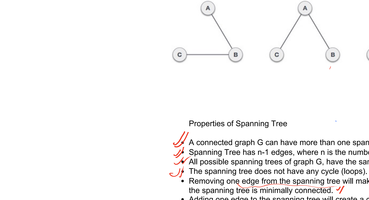


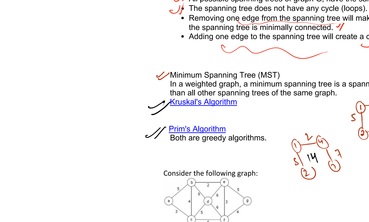


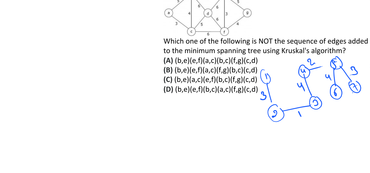






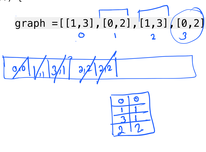


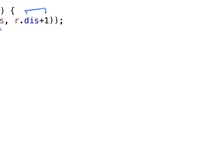


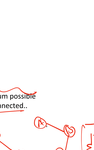


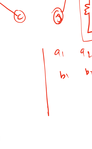

```
public boolean BFT(int[][] graph) {
        Queue<BipartitePair> q = new LinkedList<>();
        HashMap<Integer,Integer> visited = new HashMap<>();
        for (int src =0; src<graph.length; src++) {
           if (visited.containsKey(src)) {
              continue;
           }
           q.add(new BipartitePair(src, 0));
           while (!q.isEmpty()) {
              // 1. remove
              BipartitePair r = q.poll();
              // 2. Ignore if Already visited
              if (visited.containsKey()) {
                 continue;
              }
              // 3. visited marked
              visited.put(r.vtx,r.dis);
              // 4. self work
              // 5. add unvisited nbrs
              for (int nbrs :graph[r.vtx]) {
                 if (!visited.containsKey(nbrs)) {
                    q.add(new BipartitePair(nbrs, r.dis+1));
                 }
              }
           }
        }
     }
```

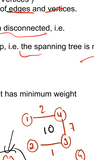
graph =[[1,3],[0,2],[1,3],[0,2]]





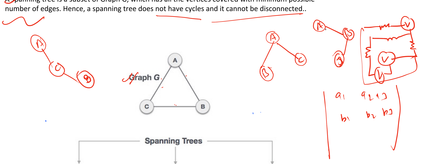


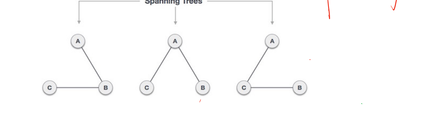


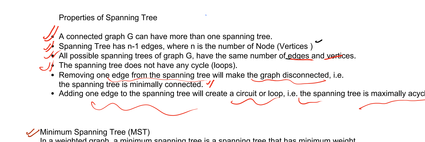


Spanning Tree

A spanning tree is a subset of Graph G, which has all the vertices covered with minimum possible number of edges. Hence, a spanning tree does not have cycles and it cannot be disconnected..

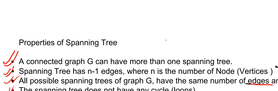


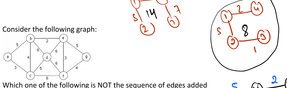




Graph G

Spanning Trees



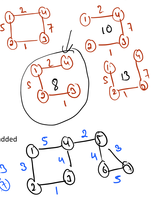


Properties of Spanning Tree

- A connected graph G can have more than one spanning tree.
- Spanning Tree has n-1 edges, where n is the number of Node (Vertices).
- All possible spanning trees of graph G, have the same number of edges and vertices.
- The spanning tree does not have any cycle (loops).
- Removing one edge from the spanning tree will make the graph disconnected, i.e. the spanning tree is minimally connected.
- Adding one edge to the spanning tree will create a circuit or loop, i.e. the spanning tree is maximally acyclic.

Minimum Spanning Tree (MST)
In a weighted graph, a minimum spanning tree is a spanning tree that has minimum weight than all other spanning trees of the same graph.
Kruskal's Algorithm

Prim's Algorithm
Both are greedy algorithms.



Consider the following graph:

Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?
(A) (b,e)(e,f)(a,c)(b,c)(f,g)(c,d)
(B) (b,e)(e,f)(a,c)(f,g)(b,c)(c,d)
(C) (b,e)(a,c)(e,f)(b,c)(f,g)(c,d)
(D) (b,e)(e,f)(b,c)(a,c)(f,g)(c,d)